

Generation Interconnection System Impact Study Report

For Queue Project Q15-0455

Lockeford 138 kV

August 27, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0455. The Interconnection Customer proposes a 125.6 MW natural gas combined cycle facility interconnecting at the Lockeford 138 kV substation in the Pennsylvania area. The request was received on April 02, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 237 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 21 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$77,326,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: LOCKEFORD / ELIZA CREEK 138kV
Contingency: ONTELAUNEE ENERGY CENTER / ELIZA CRK 345kV
Loading: 213.0% (Rating: 197 MVA)
DFAX: 49.6%
Pre-project loading: 181.3%
2. Facility: MARYMOUNT / MASSENA 138kV
Contingency: CURECANTI / MOSER GENERATING STATION 345kV
Loading: 173.3% (Rating: 248 MVA)
DFAX: 5.5%
Pre-project loading: 170.6%
3. Facility: MARYMOUNT / ENDEAVOR 138kV line
Contingency: CURECANTI / MOSER GENERATING STATION 345kV
Loading: 168.6% (Rating: 261 MVA)
DFAX: 5.5%
Pre-project loading: 166.0%
4. Facility: CURECANTI / ENDEAVOR 138kV L
Contingency: CURECANTI / MOSER GENERATING STATION 345kV
Loading: 167.6% (Rating: 268 MVA)
DFAX: 5.5%
Pre-project loading: 165.0%
5. Facility: FORT SMITH / PARKLAND SUBSTATION 138kV
Contingency: MOSER GENERATING STA / CURECANTI 345kV
Loading: 162.6% (Rating: 183 MVA)
DFAX: 5.7%
Pre-project loading: 158.7%
6. Facility: LIMERICK / MASSENA 138kV
Contingency: CURECANTI / MOSER GENERATING STA 345kV
Loading: 158.1% (Rating: 265 MVA)

DFAX: 5.5%
Pre-project loading: 155.5%

7. Facility: PARKLAND / HARVEY COUCH 138kV
Contingency: MOSER GENERATING STATION / CURECANTI 345kV
Loading: 157.4% (Rating: 182 MVA)
DFAX: 5.7%
Pre-project loading: 153.5%
8. Facility: CLODINE SUBSTATION / MARTIN CREEK 138kV
Contingency: CURECANTI / ELIZA CRK 345kV line
Loading: 153.0% (Rating: 174 MVA)
DFAX: 6.1%
Pre-project loading: 148.6%
9. Facility: KINGSBURG / FORT SMITH 138kV
Contingency: CURECANTI / MOSER GENERATING STA 345kV
Loading: 146.8% (Rating: 217 MVA)
DFAX: 5.7%
Pre-project loading: 143.5%
10. Facility: STA MVALLEY / ELK RIV SUBSTATION 138kV Line
Contingency: CURECANTI / MOSER GENERATING STATION 345kV
Loading: 145.7% (Rating: 239 MVA)
DFAX: 5.7%
Pre-project loading: 142.7%
11. Facility: STATION MVALLEY / KINGSBURG 138kV
Contingency: CURECANTI / MOSER GENERATING STA 345kV
Loading: 144.3% (Rating: 234 MVA)
DFAX: 5.7%
Pre-project loading: 141.3%
12. Facility: CLODINE / LIVINGSTON 138kV
Contingency: ELIZA CREEK / CURECANTI 345kV L
Loading: 139.7% (Rating: 205 MVA)
DFAX: 6.1%
Pre-project loading: 135.9%
13. Facility: LIVINGSTON / TRISONIC 138kV line
Contingency: CURECANTI / ELIZA CRK 345kV
Loading: 134.2% (Rating: 229 MVA)
DFAX: 6.1%
Pre-project loading: 130.9%
14. Facility: ELIZA CREEK 345/138 kV Auto
Contingency: ELIZA CREK / ONTELAUNEE ENERGY CENTER 345kV L
Loading: 128.8% (Rating: 413 MVA)
DFAX: 23.0%
Pre-project loading: 121.8%
15. Facility: MARTIN CREEK / ALLIANT SBD 9106 ROCKWELL CR 138kV Line
Contingency: CURECANTI / ELIZA CRK 345kV
Loading: 128.1% (Rating: 200 MVA)
DFAX: 6.1%
Pre-project loading: 124.3%
16. Facility: OWATONNA / MOSER GENERATING STA 345kV Line
Contingency: MOSER GENERATING STATION / LOHBUIE 345kV Line
Loading: 117.8% (Rating: 1,215 MVA)
DFAX: 17.9%

Pre-project loading: 115.9%

17. Facility: ALLIANT SBD 9106 ROCKWELL CR / DOWE FLATS 138kV

Contingency: CURECANTI / ELIZA CRK SUBSTATION 345kV

Loading: 116.5% (Rating: 212 MVA)

DFAX: 6.1%

Pre-project loading: 112.9%

18. Facility: CURECANTI / MOSER GENERATING STATION 345kV Line

Contingency: TOMBSTONE SUBSTATION / SHARYLAND 345kV L

Loading: 106.8% (Rating: 1,919 MVA)

DFAX: 19.1%

Pre-project loading: 105.6%

19. Facility: MOSER GENERATING STATION / LOCHBUIE 345kV line

Contingency: MOSER GENERATING STATION / OWATONNA 345kV

Loading: 103.7% (Rating: 1,151 MVA)

DFAX: 12.0%

Pre-project loading: 102.4%

20. Facility: CURECANTI / DOWE FLATS 138kV

Contingency: CURECANTI / ELIZA CRK 345kV Line

Loading: 103.5% (Rating: 226 MVA)

DFAX: 6.1%

Pre-project loading: 100.2%

21. Facility: ELIA CREEK SUBSTATION / S READING 138kV

Contingency: CURECANTI / ELIZA CRK 345kV

Loading: 102.3% (Rating: 264 MVA)

DFAX: 27.1%

Pre-project loading: 89.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LOCKEFORD / ELIZA CREEK 138kV Line	Rebuild line with higher-capacity conductor	\$1,903,000	100% (\$1,903,000)
MARYMOUNT / MASSENA 138kV	Rebuild line with higher-capacity conductor	\$3,034,000	100% (\$3,034,000)
MARYMOUNT / ENDEAVOR 138kV line	Rebuild line with higher-capacity conductor	\$1,545,000	100% (\$1,545,000)
CURECANTI / ENDEAVOR 138kV L	Rebuild line with higher-capacity conductor	\$8,673,000	100% (\$8,673,000)
PARKLAND / FT SMITH 138kV	Rebuild line with higher-capacity conductor	\$1,410,000	100% (\$1,410,000)
MASSENA / LIMERICK 138kV	Rebuild line with higher-capacity conductor	\$9,214,000	100% (\$9,214,000)
PARKLAND / HARVEY COUCH 138kV	Rebuild line with higher-capacity conductor	\$4,010,000	100% (\$4,010,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CLODINE SUBSTATION / MARTIN CREEK 138kV	Rebuild line with higher-capacity conductor	\$3,408,000	100% (\$3,408,000)
FORT SMITH / KINGSBURG 138kV	Rebuild line with higher-capacity conductor	\$4,850,000	100% (\$4,850,000)
STA MVALLEY / ELK RIV SUBSTATION 138kV Line	Rebuild line with higher-capacity conductor	\$9,708,000	100% (\$9,708,000)
STATION MVALLEY / KINGSBURG 138kV	Rebuild line with higher-capacity conductor	\$2,447,000	100% (\$2,447,000)
CLODINE / LIVIGSTON 138kV	Rebuild line with higher-capacity conductor	\$5,940,000	100% (\$5,940,000)
LIVINGSTON / TRISONIC 138kV line	Rebuild line with higher-capacity conductor	\$2,989,000	100% (\$2,989,000)
ELIZA CREEK 345/138 kV Auto	Replace transformer with higher-rated unit	\$12,774,000	29% (\$3,685,000)
ALLIANT SBD 9106 ROCKWELL CR / MARTIN CRK 138kV	Rebuild line with higher-capacity conductor	\$10,482,000	100% (\$10,482,000)
OWATONNA / MOSER GENERATING STA 345kV Line	Rebuild line with higher-capacity conductor	\$12,632,000	4% (\$529,000)
ALLIANT SBD 9106 ROCKWELL CR / DOWE FLATS 138kV	Rebuild line with higher-capacity conductor	\$1,520,000	100% (\$1,520,000)
CURECANTI / MOSER GENERATING STATION 345kV Line	Reconductor line	\$7,708,000	4% (\$298,000)
MOSER GENERATING STATION / LOCHBUIE 345kV line	Reconductor line	\$8,884,000	6% (\$500,000)
DOWE FLATS / CURECANTI 138kV line	Reconductor line	\$913,000	100% (\$913,000)
ELIA CREEK SUBSTATION / S READING 138kV	Reconductor line	\$268,000	100% (\$268,000)
	Total allocated to project		\$77,326,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.